

FUELS AND COMBUSTION

"Probably for only a limited number of years to come, the largest portion of the world's power will come from the combustion of fossil fuels".

1 INTRODUCTION

Fuel is a combustible substance, containing carbon as main constituent, which on proper burning gives large amount of heat, which can be used economically for domestic and industrial purposes. Wood, charcoal, coal, kerosene, petrol, diesel, producer gas, oil gas, etc. are some of the fuels.

During the process of combustion of a fuel (like coal), the atoms of carbon, hydrogen, etc. combine with oxygen with the simultaneous liberation of heat at a rapid rate. This energy is liberated due to the "rearrangement of valency elections" in these atoms, resulting in the formation of new compounds (like CO_2 , H_2O , etc.). These new compounds have less energy (or heat content) in them and, therefore, the energy (or heat) released during the combustion process is the **difference** in the energy of the reactants (C, H, and O, etc. of fuel) and that of the products formed.



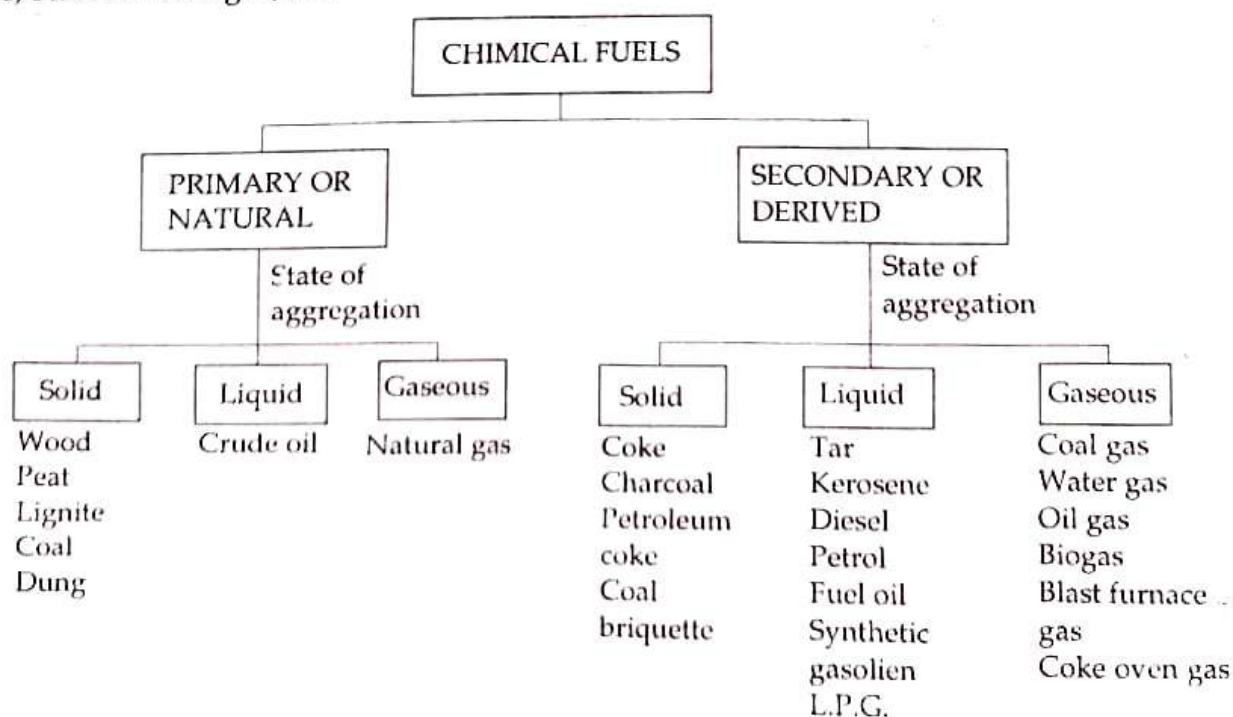
More heat energy content

Lesser heat energy content

The primary or main source of fuels are coals and petroleum oils, the amounts of which are dwindling day-by-day. These are stored fuels available in earth's crust and are, generally, called '**fossil fuels**'.

2 CLASSIFICATION OF FUELS

The fossil fuels have been classified according to their : (1) *occurrence* (and preparation), and (2) *the state of aggregation*. According to the first classification, we have : (a) **natural or primary fuels**, which are found in nature as such, e.g., wood, peat, coal, petroleum, natural gas, etc. ; (b) **artificial or secondary fuels** are those which are prepared from the primary fuels. For example, charcoal, coke, kerosene oil, diesel oil, petrol, coal gas, oil gas, producer gas, blast furnace gas, etc.



THE CHEMISTRY

The second classification is based upon their state of aggregation like : (a) solid fuels ; (b) liquid fuels and (c) gaseous fuels. The classification of fuels is summarized as on page .

Note : Colloidal fuel is a suspension finely powdered coal in fuel oil (or any other liquid fuel), generally, in the ratio of 3 : 2 by weight. Usually, some stabilizing agent is added during the preparation of colloidal fuel in order to avoid the separation of coal from the liquid fuel. In general. Such fuels possess higher calorific values and it is easy to handle them compared with powdered coal and these such fuels find applications in industrial furnaces for cement manufacture, brick firing, ceramic firing, metal processing, etc.

3 CALORIFIC VALUE

Calorific value of a fuel is "the total quantity of heat liberated, when a unit mass (or volume) of the fuel is burnt completely."

Units of heat : (1) 'Calorie' is the amount of heat required to raise the temperature of one gram of water through one degree Centigrade (15–16°C).

(2) "Kilocalorie" (or "Kilogram Centigrade units") is equal to 1,000 calories. This is the unit of metric system and may be defined as 'the quantity of heat required to raise the temperature of one kilogram of water through one degree Centigrade. Thus :

$$1 \text{ kcal} = 1,000 \text{ cal}$$

(3) "British Thermal unit" (B.Th.U.) is defined as "the quantity of heat required to raise the temperature of one pound of water through one degree Fahrenheit (60–61°F). This is the English system unit.

$$1 \text{ B.Th.U.} = 252 \text{ cal} = 0.252 \text{ kcal}$$

or
$$1 \text{ kcal} = 3.968 \text{ B.Th.U.}$$

(4) "Centigrade heat unit" (C. H. U.) is "the quantity of heat required to raise the temperature of 1 pound of water through one degree Centigrade." Thus :

$$1 \text{ kcal} = 3.968 \text{ B.Th.U.} = 2.2 \text{ C.H.U.}$$

✓ **Higher or gross calorific value :** Usually, all fuels contain some hydrogen and when the calorific value of hydrogen-containing fuel is determined experimentally, the hydrogen is converted into steam. If the products of combustion are condensed to the room temperature (15°C or 60°F), the latent heat of condensation of steam also gets included in the measured heat, which is then called 'higher or gross calorific value'. So, gross or higher calorific value (HCV) is "the total amount of heat produced, when unit mass / volume of the fuel has been burnt completely and the products of combustion have been cooled to room temperature" (i.e., 15°C or 60°F).

Lower or net calorific value : In actual use of any fuel, the water vapour and moisture, etc., are not condensed and escape as such along-with hot combustion gases. Hence, a lesser amount of heat is available. So, net or lower calorific value (LCV) is "the net heat produced, when unit mass/volume of the fuel is burnt completely and the products are permitted to escape". Alternatively, net or lower calorific value (LCV)

$$= \text{HCV} - \text{Latent heat of water vapour formed}$$

$$= \text{HCV} - \text{Mass of hydrogen} \times 9 \times \text{Latent heat of steam}$$

because 1 part by mass of hydrogen produces 9 parts by mass of water. The latent heat of steam is 587 kcal/kg or 1,060 B.Th.U./lb of water vapour formed at room temperature (i.e., 15°C).

Units of calorific value : The calorific value is, generally, expressed in *calorie/gram* (cal/g) or *kilocalorie/kg* (kcal/kg) or *British thermal unit/lb* (B.Th.U./lb) in case of solid or liquid fuel. In case of gaseous fuels, the units used are *kilocalorie/cubic metre* (kcal/m³) or *B.Th.U./cubic feet* (B.Th.U./ft³).

$$1 \text{ kcal/kg} = 1.8 \times \text{B.Th.U./lb}$$

$$1 \text{ kcal/m}^3 = 0.1077 \times \text{B.Th.U./ft}^3$$

$$1 \text{ B.Th.U./ft}^3 = 9.3 \text{ kcal/m}^3$$

...(1)

4 CHARACTERISTICS OF A GOOD FUEL

(1) **High calorific value :** A fuel should possess high calorific value, since the amount of heat liberated and temperature attained thereby depends upon the calorific value of fuel.

(2) **Moderate ignition temperature** : *Ignition temperature is the lowest temperature to which the fuel must be pre-heated so that it starts burning smoothly. Low ignition temperature is dangerous for storage and transport of fuel, since it can cause fire hazards. On the other hand, high ignition temperature causes difficulty in kindling (or igniting) the fuel, but the fuel is safe during storage, handling and transport. Hence, an ideal fuel should have "moderate" ignition temperature.*

(3) **Low moisture content** : The moisture content of the fuel *reduces the heating value* and involves in a loss of money, because it is paid for at the same rate as the fuel. Hence, *fuel should have low moisture content.*

(4) **Low non-combustible matter content** : After combustion, the non-combustible matter remains, generally, in the form of *ash or clinker*. The non-combustible matter also *reduces the heating value*, besides additional cost of storage, handling and disposal of the waste products produced. Each per cent of non-combustible matter in fuel means a heat loss of about 1.5%. Hence, *a fuel should have low content of non-combustible matter.*

(5) **Moderate velocity of combustion** : If the rate of combustion is *low*, then the required high temperature may *not* be possible, because a part of the heat liberated may get radiated, instead of raising the temperature. On the other hand, *too high* combustion rates are also *not* required.

(6) **Products of combustion should not be harmful** : Fuel, on burning, should *not* give out objectionable and harmful gases. In other words, the gaseous products of a combustion should not pollute the atmosphere. CO , SO_2 , H_2S , PH_3 , etc., are some of the harmful gases.

(7) **Low cost** : A good fuel should be readily available in bulk at a cheap rate.

(8) **Easy to transport** : Fuel must be easy to handle, store and transport at a low cost. Solid and liquid fuels can easily be transported from one place to another. On the other hand, transportation of gaseous fuels is costly and can even cause fire hazards.

(9) **Combustion should be easily controllable**, *i.e., combustion of the fuel should be easy to start or stop, when required.*

(10) **Should not undergo spontaneous combustion** : Spontaneous ignition can cause *fire hazards*.

(11) **Storage cost in bulk should be low.**

(12) **Should burn in air with efficiency, without much smoke.**

(13) In case of solid fuel, the *size should be uniform* so that combustion is *regular*.

5 COMPARISON BETWEEN SOLID, LIQUID AND GASEOUS FUELS

(A) **Solid Fuels : Advantages** : (1) They are easy to transport. (2) They are convenient to store, without any risk of spontaneous explosion. (3) Their *cost of production is low*. (4) They possess *moderate ignition temperatures*.

Disadvantages : (1) Their ash content is *high*. (2) Their large proportion of heat is wasted during combustion. In other words, their *thermal efficiency is low*. (3) They burn with *clinker* formation. (4) They combustion operations *cannot* be controlled easily. (5) Their cost of handling is *high*. (6) Their calorific value is *lower* as compared to that of liquid fuels. (7) They require *large excess of air* for complete combustion. (8) They *cannot* be used as internal combustion engine fuels.

(B) **Liquid Fuels : Advantages** : (1) They possess *higher calorific value* per unit mass than solid fuels. (2) They burn without forming dust, ash, clinkers, etc. (3) Their firing is *easier* and also fire can be extinguished easily by stopping the liquid fuel supply. (4) They are *easy to transport* through pipes. (5) They can be stored indefinitely, without *any loss*. (6) The flame produced by burning liquid fuels can easily be *controlled* by adjusting the liquid fuel supply. (7) Liquid fuels are, generally, handled by *pipes* and one man can easily regulate a large number of furnaces simultaneously. (8) They are *clean in use* and economic in labour. (9) Loss of heat to chimney is *very low*, due to greater cleanliness. (10) They require *less excess of air* for complete combustion. (11) They require *less furnace space* for combustion. (12) There is *no wear and tear* of grate bars and cleaning of fires, etc., unlike solid fuels. (13) They can be used as *internal combustion fuels*.

Disadvantages : (1) The cost of liquid fuel is relatively much *higher* as compared to solid fuels. (2) *Costly* special storage tanks are required for storing liquid fuels. (3) There is a *greater risk of fire hazards*, particularly in case of a highly inflammable and volatile liquid fuel. (4) They give *bad odour*. (5) For efficient burning of liquid fuels, *specially constructed* burners and spraying apparatus are required. (6) Choking of sprayers (during liquid fuel combustion) is a drawback of oil firing.

(C) Gaseous Fuels : Advantages : (1) They can be *conveyed easily* through pipelines to the actual place of need, thereby eliminating manual labour in transportation. (2) They can be *lighted at moment's notice*. (3) They have *high heat content* and hence, help us in having *higher temperatures*. (4) They can be *pre-heated* by the heat of hot waste gases, thereby affording economy in heat. (5) Their combustion can readily be controlled for changes in demand like oxidizing or reducing atmosphere, length of flame, temperature, etc. (6) They burn *without any soot* (or smoke) and are ashless, so there is no labour involved in ash handling, etc. (7) They are *clean* in use. (8) They do *not* require any special burners. (9) They can be produced by using even the *poorest* quantity of coal, etc. (10) They burn without heat loss, due to convection currents. (11) They burn in *slight* excess of air supply. (12) They are free from solid and liquid impurities. Hence, they do not affect the quality of metal produced, when used as a metallurgical fuel. (13) Complete combustion without pollution is possible, due to uniform mixing of air and fuel. (14) They have *high calorific values*. (15) They can also be used as *internal combustion engine fuels*.

Disadvantages : (1) Very large storage tanks are needed for them. (2) They are *highly inflammable*, so chances of fire hazards are high in their use. (3) They are *more costly* as compared to solid and liquid fuels.

6 BOMB CALORIMETER

This apparatus is used to find the *calorific value of solid and liquid fuels*.

Construction : A simple sketch of bomb calorimeter is shown in Fig. 1. It consists of a strong cylindrical stainless steel *bomb* in which the combustion of fuel is made to take place. The bomb has a *lid*, which can be screwed to the body of bomb so as to make a perfect gas-tight seal. The lid is provided with two stainless steel *electrodes* and an *oxygen inlet valve*. To one of the electrodes, a small *ring* is attached. In this ring, a nickel or stainless steel *crucible* can be supported. The bomb is placed in a copper *calorimeter*, which is surrounded by an *air-jacket* and *water-jacket* to prevent heat losses due to radiation. The calorimeter is provided with an electrically operated *stirrer* and *Beckmann's thermometer*, which can read accurately temperature difference upto 1/100th of a degree.

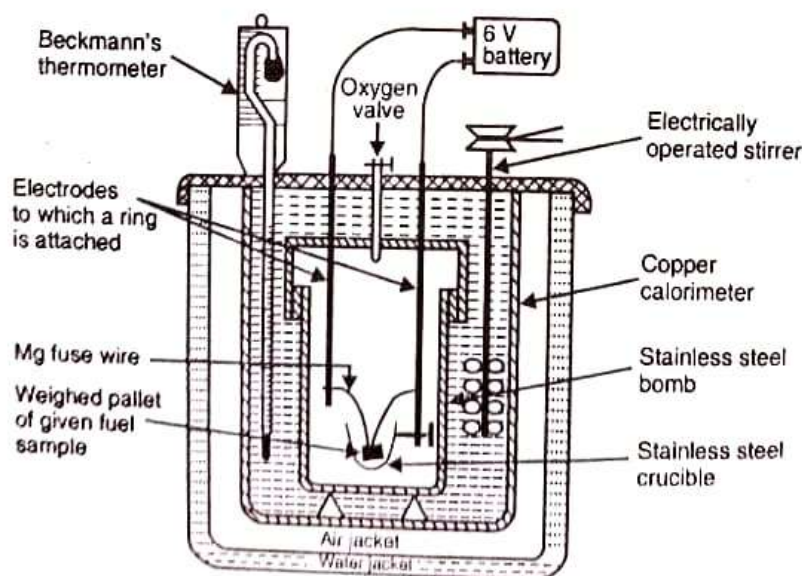


Fig. 1. Bomb calorimeter.

Working : A known mass (about 0.5 to 1.0 g) of the given fuel is taken in clean crucible. The crucible is then supported over the ring. A fine magnesium wire, touching the fuel sample, is then stretched across the electrodes. The bomb lid is tightly screwed and bomb filled with oxygen to 25 atmospheric pressure. The bomb is then lowered into copper calorimeter, containing a known mass of water. The stirrer is worked and initial temperature of the water is noted. The electrodes are then connected to 6-volt battery and circuit completed. The sample burns and heat is liberated. Uniform stirring of water is continued and the maximum temperature attained is recorded.

Calculation : Let x = mass in g of fuel sample taken in crucible ; W = mass of water in the calorimeter ; w = water equivalent in g of calorimeter, stirrer, thermometer, bomb, etc. ; t_1 = initial temperature of water in calorimeter ; t_2 = final temperature of water in calorimeter ; L = higher calorific value in fuel in cal/g.

$\therefore$ Heat liberated by burning of fuel = $x L$

and heat absorbed by water and apparatus, etc. = $(W + w) (t_2 - t_1)$

But heat liberated by the fuel = Heat absorbed by water, apparatus, etc.

$$\therefore x L = (W + w) (t_2 - t_1)$$

$$\text{or HCV of fuel (L)} = \frac{(W + w) (t_2 - t_1)}{x} \text{ cal/g (or kcal/kg)} \quad \dots(2)$$

Note : The water equivalent of the calorimeter is determined by burning a fuel of known calorific value and using the above equation. The fuels used for this purpose are benzoic acid (HCV = 6,325 kcal/kg) and naphthalene (HCV = 9,688 kcal/kg).

If H = percentage of hydrogen in fuel, then :

$$\frac{9H}{100} \text{ g} = \text{Mass of H}_2\text{O from 1 g of fuel} = 0.09 H \text{ g}$$

$\therefore$ Heat taken by water in forming steam

$$= 0.09 H \times 587 \text{ cal} \quad (\because \text{Latent heat of steam} = 587 \text{ cal/g})$$

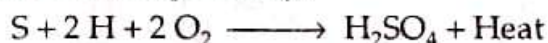
$\therefore$ LCV = HCV – Latent heat of water formed

$$= (L - 0.09 H \times 587) \text{ cal/g (or kcal/kg)} \quad \dots(3)$$

Corrections : To get more accurate results, the following corrections are applied :

(i) **Fuse wire connection :** The heat liberated, as measured above, includes the heat given out by ignition of the fuse wire used.

(ii) **Acid correction :** Fuels containing S and N are oxidised, under high pressure and temperature of ignition, to H_2SO_4 and HNO_3 respectively.



Formation of these acids are exothermic reactions. So, the measured heat also includes the heat given out during the acid formation. The amount of these acids is analysed from washings of bomb by titration ; while H_2SO_4 alone is determined by precipitation as BaSO_4 . The correction for 1 mg of S is 2.25 cal ; while for 1 mL of N/10 HNO_3 formed is 1.43 cal.

(iii) **Cooling correction :** Time taken to cool the water in calorimeter from maximum temperature to room temperature is noted. From the rate of cooling (dt°/minute) and the actual time taken for cooling (t minutes), the cooling correction of $dt \times t$ is added to the rise in temperature.

$$\therefore L = \frac{(W + w) (t_2 - t_1 + \text{cooling correction}) - [\text{Acid} + \text{fuse corrections}]}{\text{Mass of fuel (x)}} \quad \dots(4)$$